Strong Tree-Childness Is a Sharp Identifiability Boundary for Level-2 Jukes–Cantor Networks

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Abstract

We determine the generic observational equivalence relation for binary standard semi-directed strongly tree-child level-2 phylogenetic networks under the open four-state Jukes–Cantor model. A source-relative full-dimensional regular stochastic containment between two such network models occurs if and only if their labelled bridge trees agree and every pair of corresponding blobs is labelled-isomorphic up to ordinary redirection of a triangle. Thus there are no proper one-sided generic containments, and an exact generic distribution determines a unique standard semi-directed topology modulo triangle redirection. The proof is local-to-global but does not recover physical bridge lengths: Fourier cut ranks determine the bridge tree, while positive rank-one factorizations determine projective local tensors modulo the full incidence-scaling group. A finite graph-to-algebra theorem classifies all cycle and theta factors; a marginal-submersion and coherent-probe argument promotes its bounded certificates to arbitrary subdivisions. Ordinary triangle orientations share a regular local germ, but equality of their complete stochastic images is not asserted.

The boundary is sharp. For every $n \geq 4$ we give two weakly but not strongly tree-child level-2 networks that are nonisomorphic and not related by triangle redirection, yet whose open JC images share a regular relatively open subset of full dimension $2n$. The four-leaf base equality and all rank, sign, class-membership, and finite-atlas claims have exact independent replay certificates.

Keywords. phylogenetic networks; identifiability; Jukes–Cantor model; level-2 networks; strongly tree-child networks; algebraic statistics.

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1 Introduction

Phylogenetic-network identifiability asks which evolutionary histories are recoverable from an exact site-pattern distribution. Even when the substitution process on every edge is simple, reticulation choices make a network model a structured mixture over displayed trees. The central issue is therefore not merely whether one polynomial ideal differs from another, but whether two *open stochastic* model images share a region of full dimension.

For level-2 networks, two phenomena set the boundary of the problem. First, ordinary redirection of a triangle can preserve a full-dimensional local JC germ. It is therefore necessary to quotient by that move. Second, a four-leaf theta network admits a more serious pendant-transfer ambiguity in the weakly tree-child class. That pair is not strongly tree-child, so it leaves open the natural sharp question: does strong tree-childness remove every generic ambiguity other than triangle redirection?

We answer yes under the standard, reticulation-preserving semi-directed convention. Our result strengthens a triangle-free level-2 theorem of Englander et al. [5] by allowing triangles and identifying their exact quotient. It also classifies *one-sided* generic containment, which is logically stronger than pairwise equality of generic model dimensions. The result parallels the role of triangle quotients in level-1 identifiability [2], while the proof must handle the two-reticulation theta core directly. Related local and semialgebraic phenomena appear in work of Ardiyansyah [1], Cox, Gross, and Martin [3], Currie et al. [4], and Sullivant [9].

The proof has four layers. A pointwise Fourier flattening theorem recovers all bridge splits even under one-sided containment. Rank-one cut factorizations then recover each local boundary tensor only projectively; the correct fiber is the full incidence-scaling action, not one reciprocal scalar per bridge. A structural generator theorem reduces every level-2 blob to a cycle or one of four directed theta cores. Finally, exact graph-derived invariants classify a rigid bounded support of every core, and a submersion plus overlapping one- and two-port probes reconstructs arbitrary subdivision words. This finite step is computer-assisted, but its theorem object is a decorated directed relation carrying both graphs, all port matches, all incoming roles, and a complete graph-to-polynomial provenance chain.

Our two principal results are as follows.

Theorem 1.1 (Strong-class classification)**.** *Let N, N' be leaf-labelled binary standard semi-directed strongly tree-child level-2 networks on the same taxon set. Under the open Jukes–Cantor model,*

$$N \preceq_{\mathrm{JC}} N' \iff \text{\textit{their labelled reduced bridge trees agree, and every pair of corresponding nontrivial blobs is labelled-isomorphic or differs by ordinary triangle redirection } \mathsf{T}.}$$

Consequently there is no proper one-sided generic containment. Moreover, $N \bowtie_{\mathrm{JC}} N'$ holds exactly under the same condition.

Corollary 1.2 (Generic identifiability modulo triangles)**.** *For every fixed N in the class there is a proper algebraic subset $E_N \subsetneq \mathcal{V}_N^{\mathrm{JC}}$ such that every distribution $p \in \mathcal{M}_N^{\mathrm{JC}} \setminus E_N$ determines the labelled standard semi-directed topology of N modulo ordinary triangle redirection.*

Theorem 1.3 (Sharpness)**.** *For every $n \geq 4$ there are two leaf-labelled binary standard semi-directed level-2 networks $K_n, K'_n \in W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$ such that:*

- *(i) they are nonisomorphic and not related by ordinary triangle redirection;*
- *(ii) each open JC model image has dimension $2n$; and*
- *(iii) their images share a regular relatively open subset of dimension $2n$.*

Thus strong tree-childness is a sharp boundary against the full weakly tree-child class. The qualifier "generic" in corollary 1.2 means the complement of a proper algebraic exceptional set. The local triangle statement is different: we prove one common full-dimensional regular Euclidean-open germ, not equality or density of the complete stochastic images.

2 Conventions and observational relations

2.1 Rooted and semi-directed networks

Let X be a finite set of taxon labels. An *LSA-valid rooted binary phylogenetic network* on X is a finite directed acyclic graph without parallel arcs in which the root has bidegree $(0, 2)$, a tree vertex has bidegree $(1, 2)$, a reticulation has bidegree $(2, 1)$, and each leaf has bidegree $(1, 0)$ and a unique label in X. Every vertex is reachable from the root. The root is the lowest stable ancestor: it lies on every root-to-leaf path and no proper descendant does so for every labelled leaf. A rooted presentation is *tree-child* if every internal vertex has a tree vertex or leaf as a child.

We use one reticulation-preserving semi-deorientation, denoted sd_0. Mark every arc entering a reticulation, undirect every other arc, delete the binary root, and replace its two incident edges by one edge between its children, retaining at either endpoint exactly the arrowhead present before deletion. A presentation is admissible only if this single suppression produces a simple binary mixed graph: no loop or parallel edge is created and every reticulation and arrowhead survives. No later degree-two or parallel cleanup is part of sd_0. Isomorphisms preserve labels, undirected edges, retained arrowheads, and vertex roles. An *admissible rooting* of a mixed graph is an LSA-valid rooted binary network whose sd_0 image is exactly that graph. This is the already-simple convention used by Englander et al. [5] and the binary LSA-valid specialization of Holtgrefe et al. [7]; it is not a broader exhaustive cleanup after taking an induced subnetwork.

We distinguish

$$
\begin{aligned}
R_{\mathrm{TC}} &= \{\text{chosen rooted tree-child presentations}\},\\
W_{\mathrm{TC}} &= \{\text{mixed graphs with at least one tree-child admissible rooting}\},\\
S_{\mathrm{TC}} &= \{\text{mixed graphs with at least one admissible rooting, all of which are tree-child}\}.
\end{aligned}
$$

A simple binary mixed graph with an admissible rooting lies in S_{TC} exactly when every tail of a retained reticulation edge is incident with two undirected edges. Equivalently, it has no *omnian* tail. This excludes both a tree vertex with two reticulation children and a reticulation with a reticulation child. Conversely, such a tail has one ordinary incidence as parent and the other as a nonreticulate child in every admissible rooting; vertices with no outgoing reticulation edge and the inserted root satisfy the tree-child condition as well.

A cut edge is an edge whose deletion disconnects the underlying graph. A *blob* is a maximal connected subgraph containing no cut edge. In our simple binary setting, nontrivial blobs are the usual biconnected blocks. A network is *level two* when every blob contains at most two reticulations. Contracting each nontrivial blob and retaining ordinary trivalent components gives the reduced, leaf-labelled *bridge tree*.

Definition 2.1 (Ordinary triangle redirection)**.** *An ordinary triangle redirection* T *changes only which vertex of a triangle is reticulate and the two arrowheads entering it. It leaves the labelled underlying graph, every pendant-component placement, and every arrowhead outside the triangle unchanged.*

The generated local relation is denoted $\sim_{\mathsf{T}}$. Both endpoint topologies in any invocation below are required to belong to the theorem's class.

2.2 The open JC model

Identify the four states with $G = \mathbb{Z}_2 \times \mathbb{Z}_2$. Every edge e has a single nontrivial Fourier multiplier $x_e \in (0,1)$, so its transition matrix is

$$T_e(a,b) = \begin{cases} (1+3x_e)/4, & a = b, \\ (1-x_e)/4, & a \neq b. \end{cases}$$

Every reticulation r has an inheritance probability $\lambda_r \in (0,1)$, and the root state is uniform. Conditional on choosing one parent at every reticulation, the network becomes a displayed tree. For $g = (g_i)_{i\in X} \in G^X$,

$$\widehat{p}_N(g) = \begin{cases} \sum_\sigma w_\sigma \prod_{e\in T_\sigma} x_e^{\mathbf{1}[\oplus_{i\in A_e} g_i \neq 0]}, & \oplus_{i\in X} g_i = 0, \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where $A_e \mid A_e^c$ is the displayed-tree split. This is the usual group-based Fourier formula [6, 8].

Rerooting along ordinary edges does not change (1): every displayed unrooted tree and split is unchanged, and complementing a parent choice only replaces λ by $1-\lambda$. Thus the JC model of the standard semi-directed topology is independent of its admissible rooted presentation. Write $\mathcal{M}_N^{\mathrm{JC}}$ for its open stochastic image and $\mathcal{V}_N^{\mathrm{JC}}$ for the complex Zariski closure.

Definition 2.2 (Observational relations)**.** *We write $N \bowtie_{\mathrm{JC}} N'$ if there is an interior distribution regular for both parameterizations and a relative neighborhood of that point in the intersection having the full local dimension of both images. We write $N \preceq_{\mathrm{JC}} N'$ if a relatively open subset of the full local source model lies in the target image; the target dimension is allowed to be larger.*

These relations are distinct from equality of Zariski closures, equality of complete stochastic images, lower-dimensional intersection, and boundary-only intersection.

3 Primitive level-2 factors

Cut away the components beyond the boundary edges of one blob, retain their attachment sites as labelled ports, and contract every ordinary port-free bivalent path.

Proposition 3.1 (Primitive-core theorem)**.** *Every nontrivial factor obtained from a standard binary level-2 blob has a cycle core or one of four directed theta cores. Every full factor is recovered uniquely by finite ordered words of port-bearing subdivisions on the directed core segments and by one real child port below every path-sink reticulation.*

Proof. Choose an admissible rooting and expose the incoming boundary. If the factor has t tree vertices, r reticulations, v internal vertices, and e internal edges, summing internal indegrees gives

$$e = t + 2r - 1 = v + r - 1.$$

Its cyclomatic number is therefore r. Rank one gives a cycle. At rank two, every vertex surviving port-free degree-two suppression has degree three. Since $e = v + 1$, we have $3v \leq 2e = 2v + 2$, so $v \leq 2$; biconnectedness and simplicity leave two poles joined by three paths, namely a theta.

Direction changes along a theta path only at a source event or at a reticulation-sink event. Both poles cannot be reticulations, for their outgoing pole-to-pole paths would make a directed cycle. If one pole is reticulate, the internal source and the second reticulation sink lie on the same path or on distinct paths. If both poles are tree vertices, the source shares a path with one of the two internal sinks or lies on the third path. These are the four cores. Reading port vertices in directed order on each segment recovers the words, and contracting those words recovers the unique core. □

For reference, the segment templates and minimum no-omnian repairs are:

core	directed segments	minimum repairs
cycle	$S \to X, S \to X$	$\{0\}, \{1\}$
θ_0	$S \to U, S \to V, U \to X, V \to X, U \to V$	$\{2,3\}, \{3,4\}$
θ_1	$S \to U, S \to X, V \to X, U \to V, U \to V$	$\{2,3\}, \{2,4\}$
θ_2	$S \to U, S \to V, U \to X_0, V \to X_0, U \to X_1, V \to X_1$	$\{2,3\}, \{2,5\}, \{3,4\}, \{4,5\}$
θ_3	$S \to U, S \to X_0, V \to X_0, U \to X_1, V \to X_1, U \to V$	$\{2\}, \{4\}$

Occupying one listed repair is necessary and sufficient because occupancy is monotone and the repairs respectively give a nonreticulate child to each offending tail, separate a reticulation child, or split a parallel core pair.

Proposition 3.2 (Triangles are automatically unique)**.** *Every simple standard semi-directed network in S_{TC} has at most one triangle in each level-2 blob.*

Proof. A cycle has one simple cycle. A theta with path lengths (ℓ_1, ℓ_2, ℓ_3) has cycle lengths $\ell_i + \ell_j$. Two triangles force $(1,1,2)$ or $(1,2,2)$. The first has parallel pole edges. The second is $K_4 - e$. Two reticulations require four incoming arrowheads. A reticulation cannot tail one of these arrowheads, and the no-omnian condition allows each ordinary vertex to tail at most one. The mixed core has only two ordinary vertices. Before root suppression, the inserted root supplies at most one additional compatible tail, still fewer than four. Hence no admissible rooting is tree-child. □

Thus theorem 1.1 applies to all of S_{TC} at level two; no separate one-triangle-per-blob assumption is needed.

Lemma 3.3 (Root reduction)**.** *Every root-containing S_{TC} factor has the same open projective JC tensor germ as an incoming-port presentation at some real leaf-bearing boundary. For a comparison, the source and target incoming boundaries may be chosen independently.*

Proof. Starting at an admissible root, repeatedly choose a tree or leaf child. This reaches a labelled leaf without crossing a reticulation. Move the root down that ordinary path to the associated real boundary arm. Only ordinary arcs reverse. Every former parent on the path becomes a tree child, retained arrowheads stay fixed, and no directed cycle is created. The two root sides both contain a leaf, since otherwise the original root was not the LSA. The new site is admissible and is tree-child because membership in S_{TC} quantifies over every admissible rooting.

JC reversibility preserves all displayed unrooted trees. Splitting an arm multiplier x into $x_1 x_2 = x$ or multiplying it after suppression stays inside $(0, 1)$. The only tensor change from moving within the arm is a positive incidence factor. Suppressing the moved root returns the same mixed graph. The construction is existential for each factor and supplies no reason that two factors share one incoming boundary. □

4 Pointwise recovery of cuts

For a split $A \mid A^c$, arrange the Fourier flattening by the total character $h = \oplus_{i\in A} g_i$. It has four character blocks.

Theorem 4.1 (Pointwise cut theorem)**.** *At every point of the open JC model of a network in theorem 1.1,*

$$A \mid A^c \text{ is a cut split} \quad \Longleftrightarrow \quad \operatorname{rank} \mathrm{Flat}_{A|A^c}(p) \le 4.$$

Proof. At a cut, each character block is a positive rank-one outer product of the two side tensors, with the bridge multiplier appearing in nonzero sectors. Hence the rank is at most four.

For the converse, a noncut has either one active central factor or a central bridge joining two active three-port endpoint tensors. The one-active cases reduce, after word compression, to the finite exact strict-minor theorem in lemma 4.2 below. For the two-active case write

$$a = P(1,1,0), \quad b = P(1,0,1), \quad c = P(0,1,1), \quad t = P(1,2,3)$$

and use capitals for the second endpoint. Set $F = abc - t^2$ and $G = a - bc$. Every allowed endpoint satisfies

$$F > 0 \quad \text{or} \quad F = 0 \text{ and } a \ge bc. \tag{2}$$

The ordinary trivalent endpoint is the equality case. If the endpoints are joined by $0 < z < 1$, rank one in all four crossing blocks forces the vanishing of minors including

$$f_1 = aA - z^2 bcBC, \qquad f_2 = zTt - z^2 bcBC,$$
$$f_3 = zC(At - zTbc), \qquad f_4 = zc(zBCt - Ta).$$

The exact identities

$$aA - zTt = f_1 - f_2,$$
$$z^2 CT(abc - t^2) = zCt(f_1 - f_2) - af_3,$$
$$z^2 ct(ABC - T^2) = Af_4 + zcT(f_1 - f_2)$$

then force $F = F' = 0$. By (2), $aA \ge bcBC > z^2 bcBC$, contradicting $f_1 = 0$.

For arbitrary words, take one representative from every monochromatic run of a proposed split on each directed segment, duplicating an adjacent representative when balance requires it. Occupied repair segments remain occupied. A split displayed by every full switching would remain displayed after this restriction; the exact bounded theorem has no noncut survivor. Thus a block minor is strictly nonzero at every open point. □

Lemma 4.2 (Exact cut certificate)**.** *The endpoint dichotomy* (2) *and every one-active case used in theorem 4.1 have graph-derived exact certificates. The independent replay contains* 67 *endpoint types with* $F > 0$, 9 *with* $F = 0$ *and* $G > 0$*, the ordinary endpoint with* $F = G = 0$*, and* 204 *strict one-active minors. All inequalities hold throughout the complete open parameter cube.*

Proof. For every primitive completion the verifier enumerates the displayed switchings, computes descendant masks, forms the exact Fourier tensor, and pulls back F, G or the selected flattening minor. Each nonzero pullback is factored or converted to the Bernstein basis on the unit cube; all coefficients required for strict sign have one sign. A second implementation reconstructs all graph-to-polynomial assignments without importing the producer. The full normalized records and all 204 minors agree. This is a finite exact proof because proposition 3.1 and word compression exhaust the endpoint types; the counts are checksums, not the premise. □

Corollary 4.3 (Bridge-tree equality under one-sided containment)**.** *If $N \preceq_{\mathrm{JC}} N'$, then N and N' have the same labelled cut splits and the same reduced bridge tree, including ordinary/nontrivial component decoration.*

Proof. On the common source-open set, a source cut cannot be a target noncut by theorem 4.1. Conversely, the rank equations for a target cut hold on the same set and cannot hold at any source noncut point. Hence both cut sets are equal. Their compatible split system reconstructs the reduced labelled tree. For a degree-three component, $F = abc - t^2$ is zero on an ordinary median and strictly positive on an open three-sunlet. A strong theta has at least four boundaries. Thus component decorations agree as well. □

5 Projective bridge peeling

Let T be the recovered bridge tree. A component tensor P_v is supported on assignments satisfying local character conservation. For a bridge $e = uv$ with multiplier x_e, the global Fourier tensor is

$$\Gamma(P, x)(g) = \prod_{v} P_v(g_v, h_v) \prod_{e \in E(T)} x_e^{\mathbf{1}[h_e \neq 0]}. \tag{3}$$

Theorem 5.1 (Exact positive bridge fiber)**.** *On the positive JC-symmetric locus, the complete fiber of (3) is the incidence action*

$$P_v \longmapsto P_v \prod_{e \ni v} a_{v,e}^{\mathbf{1}[h_e \neq 0]}, \qquad x_e \longmapsto \frac{x_e}{a_{u,e} a_{v,e}}, \tag{4}$$

with all $a_{v,e} > 0$. Every retained component admits a positive real-analytic local slice. The sliced component tensors and normalized effective bridge scales are intrinsic functions of the distribution.

Proof. Cut T at one bridge and fix a character sector h. The corresponding flattening block is a positive rank-one matrix. Uniqueness of positive rank-one factors gives one scale on either side. The all-zero entry fixes the zero-sector scale, and the six automorphisms of G act transitively on the three nonzero characters, so JC symmetry gives one nonzero scale per incidence. Peeling from the leaves of T yields exactly (4); direct cancellation proves the converse.

For a component carrying a physical leaf block, one-character anchor ratios normalize the incidences independently. An unmarked retained component has degree at least three; pair anchors $(1, 2), (1, 3), (2, 3), (1, 4), \ldots$ have a full-rank log-exponent matrix. Positive square-root formulas give analytic slices. The excluded degree-two stabilizer would require the non-strong $K_4 - e$ factor. Because T is a tree, no gauge circulates around it. □

Remark 5.2. The old reciprocal-only transformation is false. For example, $(y, z, x) = (1/2, 1/2, 1/2)$ and $(3/5, 3/5, 25/72)$ have the same observable product $xyz = 1/8$ but are not related by reciprocal endpoint scaling. The theorem recovers normalized effective scales, not physical bridge multipliers.

Proposition 5.3 (Localization and no compensation)**.** *If $N \preceq_{\mathrm{JC}} N'$, then for every pair of corresponding nontrivial factors H, H' there is a source-relative full-dimensional regular containment of the projective local JC germ of H in one projective incoming/completion germ of H'. Parameters in other blobs cannot compensate for a local separator.*

Proof. In source slice coordinates, the common source-open set contains a smaller product box of local tensor germs and effective bridge intervals. Vary one focal factor and fix the other coordinates. Every target realization of the same distribution has the same intrinsically extracted focal projective orbit by theorem 5.1. The finitely many target incoming and completion types cover the focal box. Semialgebraic dimension therefore places a source-relative full-dimensional open *subgerm* into one type. No continuous choice of target parameters is made. Since extraction is an intrinsic function of the distribution, a distant factor cannot change the focal orbit or cancel a projective invariant. ☐

6 The complete local theorem

For a port tensor $P(g_1, \dots, g_k)$, quotient by the positive action

$$P(g) \longmapsto P(g) \prod_i a_i^{\mathbf{1}[g_i \neq 0]}.$$

Let $\mathbb{P}\mathcal{M}_H$ denote the resulting open projective JC model of a complete ported factor H.

Theorem 6.1 (Local blob-containment theorem)**.** *Let H, H' be complete nontrivial factors on the same labelled physical boundary set, induced by standard semi-directed S_{TC} level-2 networks. Choose admissible incoming presentations independently. Then*

$$\mathbb{P}\mathcal{M}_H \preceq \mathbb{P}\mathcal{M}_{H'} \quad \Longleftrightarrow \quad H \textit{ and } H' \textit{ are labelled-isomorphic or } \mathsf{T}\textit{-related.}$$

There is no proper one-sided projective local containment.

The necessity proof is finite after a structural support reduction. We give its mathematical completeness argument before stating the exact computer-assisted lemma.

6.1 Rigid supports and exact completions

Choose every reticulation-sink child and one labelled port on each segment of one minimum repair. The resulting support Q retains the primitive core and is pointwise rigid. Its outgoing sizes are

$$2 \quad (\text{cycle}), \qquad 3 \quad (\theta_0, \theta_1, \theta_3), \qquad 4 \quad (\theta_2).$$

The minimal two-outgoing cycle is handled directly by

$$F = q_{011} q_{101} q_{110} - q_{123}^2, \tag{5}$$

which vanishes on an ordinary median and is strictly positive on every open three-sunlet. The three sunlet orientations are exactly T.

Marginalize a hypothetical full containment to Q. On every target core, distribute the selected ordinary labels in order among its directed segments. Its structural incoming boundary may be selected or have character zero. Restore every omitted reticulation-sink child and every empty segment of a chosen minimum repair by one zero-character dummy. Any further omitted ordinary port lies in a serial class and contributes only through a positive product. Conversely, contracting unselected ordinary subdivisions in a full strong target produces exactly one such raw record. Hence the completion grammar is onto every selected target tensor.

For each displayed switching and physical edge, record the descendant mask on the selected ports. Edges having the same complete row of masks occur in every Fourier monomial with equal exponents and are replaced by their product. The map

$$(x_1, \dots, x_s) \longmapsto x_1 \cdots x_s$$

is onto $(0,1)$ and has nonzero differential there. Parent flips replace λ by $1-\lambda$. Thus the descriptor gives the exact positive selected tensor image, not a boundary specialization.

On every ordered four-port marginal the certificate evaluates the S_4-orbit of seven explicit integer JC invariant templates. The templates and all 84 distinct transported polynomials are in the machine-readable appendix. Every polynomial is multihomogeneous in the port arms, so its zero or strict-sign status descends to the projective quotient. If an invariant vanishes identically on the target but not on the source, source-full containment would put an open part of an irreducible source model in a proper algebraic subset. This gives the exact necessary filter

$$S(\text{source}) \subseteq S(\text{target}),$$

where S records nonidentically-zero pullbacks. A modular nonzero value is used only as an exact nonvanishing witness; every modular zero is expanded over $\mathbb{Z}$.

6.2 The bounded exact theorem

Lemma 6.2 (Graph-to-algebra bounded classification)**.** *Every decorated directed relation surviving the necessary invariant filter is either:*

(a) strictly separated on the open cube by a regenerated polynomial or flattening minor;
(b) a nonretaining relation bound to one exact fixed-full restoration forest, all of whose terminal paths are separated or labelled-isomorphic modulo T*; or*
(c) directly labelled-isomorphic modulo T*.*

No nonisomorphic, non-T *relation survives, in either containment direction.*

Proof. The theorem object is the coloured disjoint union of the source and target mixed graphs together with side colours, independently distinguished incoming boundaries, all physical port labels and matches, omitted-role placeholders, and source-to-target direction. Canonicalization retains explicit vertex, edge, inheritance, label, and Fourier-coordinate transports. Thus two relations sharing an unlabelled target cannot be merged by a target hash.

For the three-outgoing supports, independent generation gives 10,466 canonical relations from 10,826 raw presentations. There are 5,284 strict relations, represented by 800 distinct exact polynomial bodies. The 5,120 pending canonical relations have 5,344 raw coverages, bijective with all 5,344 restoration roots. The restoration forest has 68,584 states: 56,055 generic polynomial separations, 8,349 refinements, 4,036 strict separations, 120 labelled-isomorphism terminals, and 24 T terminals. The 62 direct residual relations independently reduce to 34 labelled isomorphisms and 28 ordinary T relations.

The four-outgoing θ_2 gate independently generates all three source supports, both target incoming modes, 6,138 completion records, every relative five-port assignment, and the full 84-invariant deck. Exactly three necessary signature pairs survive, all equal. Expanding provenance gives 192 raw presentations, intrinsically partitioned into

$$\begin{aligned} 18 &\text{ direct isomorphisms,} \\ 42 &\text{ selected-incoming rooting duplicates,} \\ 132 &\text{ marginalized-incoming restoration roots.} \end{aligned}$$

An independent mixed-graph canonicalizer gives explicit transports for all 42 duplicates and proves that the 132 marginalized records equal the frozen root multiset exactly. Their restoration forest has 2,106 states: 1,860 generic separations, 114 refinements, and 132 labelled isomorphism terminals.

For every strict record the verifier reconstructs both rooted graphs, enumerates displayed switchings and descendant masks, forms the complete exact JC tensors, regenerates the chosen pullback, and proves its open-cube sign by exact factors or rational Bernstein coefficients. For every equality terminal, independent standard mixed-graph canonicalization accepts only labelled isomorphism or the quotient forgetting the arrowheads of one triangle. Complete normalized primary and clean-room records agree.

The finite universe is exhaustive by proposition 3.1, the onto completion grammar, every relative port permutation, and both incoming modes. The integer counts above are reproducibility checksums. Mutation suites reject relation deletion or duplication, changed port correspondences, collapsed source embeddings, swapped separators, reversed direction, changed incoming roles, broken restoration parents, and valid polynomials attached to the wrong graph. Therefore the exact finite alternatives (a)–(c) hold. □

6.3 Restoration and arbitrary words

The hard cover never infers a larger marginal from containment of a smaller one. Fix the complete relation first. Let Q_s, Q_t be source and target rigid supports, and set $A = Q_s \cup Q_t$. Every target dummy role is filled by its actual physical label in the full comparison. For each prefix D_j of those labels,

$$H|_{Q_s \cup D_j} \preceq H'|_{Q_s \cup D_j}$$

is a direct marginal of the original full containment. On the source side, the restriction map to effective serial products is an open semialgebraic submersion. Intersecting with the generic constant-rank locus shows that each prefix contains a source-relative open selected-model subgerm. Hence the actual restoration path cannot meet a separating certificate and must terminate in a common core-retaining anchor A modulo T.

The 62 direct residual anchors require a separate base case: they have four selected ports, while every restoration terminal has at least five. Each has one canonical transport (34 labelled isomorphisms and 28 ordinary T relations). Independently inserting one new labelled port on every ordered source–target pair of admissible blob arcs gives all 2,642 $A \cup \{p\}$ relations. Exactly 314 remain isomorphic or T-related. Inserting a second port above precisely those parents gives all 18,224 $A \cup \{p, q\}$ relations, of which 2,032 remain isomorphic or T-related. Every other relation has a regenerated exact JC separator, and every surviving child transport restricts its unique parent transport. Thus direct anchors and restoration terminals satisfy the same coherence hypothesis below.

Lemma 6.3 (Coherent-probe theorem)**.** *After fixing the one anchor transport on A, the restrictions $A \cup \{p\}$ locate every additional port, and $A \cup \{p, q\}$ determines the order of each pair on one segment. All probes assemble to one complete port-word isomorphism modulo one coherent ordinary triangle redirection. At most ten tensor ports are required.*

Proof. The anchor is pointwise rigid on both sides. Therefore every one-port probe restricts the same core map and fixes the segment and interval containing p. Every two-port probe restricts its exact one-port parent and determines the order of p, q when they lie in one interval. These comparisons are restrictions of actual finite total words and assemble uniquely. If a probe subdivides a triangle edge, its restriction has no triangle and forces literal orientation agreement; otherwise the same unique triangle persists, so all probes have one common T choice.

The source and target rigid supports together use at most eight tensor ports in the attained relations; adding two probes gives ten. Independent compact path-bound replay covers 101,148 three-outgoing and 168,582 θ_2 probe relations. The independently compiled direct-anchor package covers the additional $2{,}642 + 18{,}224$ relations. Exact adversarial word tests include empty segments, repeated subdivisions, symmetric labels, reversed orders, and inconsistent T choices. Every mutation is rejected. □

The left-to-right implication of theorem 6.1 now follows from lemmas 6.2 and 6.3.

6.4 The ordinary triangle converse

Lemma 6.4 (Ordinary triangle germ)**.** *The three labelled orientations of a port-labelled three-sunlet share a four-dimensional regular open JC tensor germ in the strict parameter domain. The same is true after inserting the triangle in any unchanged corresponding local context.*

Proof. At an exact rational point, direct displayed-tree summation gives equality of all 64 Fourier entries for all three orientations. Their four nonconstant JC orbit coordinates are

$$(q_{12}, q_{13}, q_{23}, q_{123}) = \left(\delta^2, \delta^2, \delta^2, \frac{4}{5}\delta^3\right), \qquad \delta = 2^{-10}.$$

Each physical orientation has a nonzero rank-four Jacobian minor. Four is also the normalized three-port ambient dimension, so every orientation maps submersively onto a neighborhood of the common tensor. Positive port-arm scaling preserves a common projective germ.

Let $f_i : \Theta_i \to \mathcal{Q}$ be the normalized triangle-tensor map for orientation i. Since df_i has rank four, after shrinking about the certified point the constant-rank theorem gives a common open tensor neighborhood U and analytic local sections $s_i : U \to \Theta_i$. For the unchanged external context let $\Phi(Q, C)$ be the common tensor contraction, and let d be its generic rank. A d-minor of $D\Phi$ is a nonzero analytic function on the connected tensor–context chart, so it cannot vanish on the nonempty open set $U \times \mathcal{C}$. Choose (Q, C) in that set where the minor is nonzero. The chain rule at $(s_i(Q), C)$ shows that both physical blob parameterizations have rank d there. Shrinking once more, $\Phi(U \times \mathcal{C})$ is a common d-dimensional regular germ, equal to the full local model dimension on both sides. This is a forward contraction, not an attempted inverse factorization of a theta into two hidden-pole pieces. □

Labelled isomorphism is immediate, and lemma 6.4 proves the converse direction of theorem 6.1. Notice that the necessity argument classified the complete theta tensor directly; it did not treat a triangle-bearing theta as a bridge gluing of a sunlet and its complementary path.

7 Proof of the global theorem

Proof of theorem 1.1. Assume $N \preceq_{\mathrm{JC}} N'$. By theorem 4.1, their labelled decorated bridge trees agree. The exact bridge fiber and proposition 5.3 induce a source-relative projective containment for every pair of corresponding nontrivial factors. Root factors reduce to independently chosen real incoming ports by lemma 3.3. The local theorem then forces each pair to be labelled-isomorphic or T-related.

Conversely, choose the identical projective germ for an isomorphic pair and the common germ of lemma 6.4 for a T pair. Shrink all finitely many germs so that both networks have positive analytic incidence representatives. For each bridge $e = uv$, choose a common effective scale z_e in a sufficiently small positive interval and put

$$x_e^{(k)} = \frac{z_e}{a_{u,e}^{(k)} a_{v,e}^{(k)}}.$$

Both physical multipliers then lie in $(0, 1)$. The bridge graph is a tree, so different bridge choices are independent and there is no scaling holonomy. The product of local common germs and bridge intervals contracts to the same global distributions in both networks. The analytic extraction map is its local inverse, so dimensions add and the common set has the full regular dimension of each model. This proves both the converse containment and $\bowtie_{\mathrm{JC}}$ statement. Since the right-hand condition is symmetric, no proper one-sided containment occurs. $\square$

Proof of corollary 1.2. For fixed n there are finitely many topologies in the class. Tree-child paths give $r \leq n - 1$ reticulations, binary degree counting gives $t = n + r - 2$, and a rooted presentation has at most $4n - 3$ vertices.

For a fixed topology N, put $d_N = \dim \mathcal{V}_N^{\mathrm{JC}}$. Its complex model closure is irreducible as the closure of a polynomial image of an irreducible parameter space. Let

$$E_{\mathrm{top}}(N) = \bigcup_{N' \not\sim_{\mathsf{T}} N} \overline{\mathcal{M}_N^{\mathrm{JC}} \cap \mathcal{M}_{N'}^{\mathrm{JC}}}^{Z, \mathcal{V}_N}.$$

Every member is proper. Otherwise its semialgebraic intersection would have full real dimension on a regular source stratum and hence nonempty relative interior, giving $N \preceq_{\mathrm{JC}} N'$ contrary to theorem 1.1. Enlarge this finite union by the singular locus and by the Zariski closures of images of parameter loci on which the Jacobian rank is less than d_N. Each latter image has dimension at most $d_N - 1$ in characteristic zero. Finally adjoin the zero sets of the observable Fourier cut anchors and atlas-witness polynomials used below; every such polynomial is nonidentically zero on its intended regular stratum. The resulting E_N is still proper. No unspecified physical-parameter factor is projected to distribution space. Outside E_N, every realizing topology is T-equivalent to N. $\square$

8 Structural reconstruction

The proof gives an exact terminating reconstruction algorithm for a generic distribution known to arise in the class.

R1. Fourier-transform the pattern probabilities.

R2. Test split flattenings using theorem 4.1; recover the compatible cut system and build the labelled reduced bridge tree.

R3. Use (5) to distinguish ordinary degree-three components from three-sunlets.

R4. Factor the positive rank-one cut blocks and impose the analytic incidence normalizers from theorem 5.1; recover every projective local tensor orbit and normalized effective bridge scale.

R5. Evaluate the bounded graph-derived invariant deck and restoration certificates to identify one rigid support of each nontrivial factor modulo T.

R6. Use common-anchor one- and two-port probes to recover every segment and every total port order.

R7. Put each triangle in the lexicographically least labelled T-orientation and assemble the canonical mixed graph.

Every loop is finite and every local probe uses at most ten tensor ports. A direct implementation uses at most $2^{n-1} - 1$ split tests and $O(n^{10})$ bounded local probes, excluding the bit complexity of exact algebraic-number operations. Thus we make no unproved polynomial-in-n claim, while the combinatorial operation count is polynomial in the length of the explicit 4^n input table. The algorithm returns the canonical topology modulo T; it does not recover physical bridge multipliers. To list which orientations realize one particular distribution, one must test open-model membership for the finite T variants. The common-germ theorem does not imply complete stochastic-image equality.

9 Sharpness outside the strong class

We now prove theorem 1.3. The argument is included to make the boundary theorem self-contained; its exact verifier is also released as a frozen independent package.

9.1 The four-leaf pair

Let the internal vertices be a root ρ, tree vertices A, B, D, E, and reticulations C, F. Both rooted networks have arcs

$$\rho \to A, \quad \rho \to C, \quad A \to B, \quad B \to C, \quad C \to D, \quad D \to E, \quad A \to F, \quad E \to F. \tag{6}$$

Their pendant arcs are

$$\begin{aligned} K_4 : &\quad B \to 1,\ D \to 2,\ F \to 3,\ E \to 4, \\ K_4' : &\quad E \to 1,\ D \to 2,\ F \to 3,\ B \to 4. \end{aligned}$$

After root suppression, the nontrivial blob is the theta with A–C paths

$$A - C, \qquad A - B - C, \qquad A - F - E - D - C,$$

whose cycle lengths are $3, 5, 6$.

Proposition 9.1 (Combinatorics of the weak pair)**.** *The two reductions lie in $W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$, are level two, and are neither isomorphic nor T-equivalent.*

Proof. The displayed rooted networks are binary, acyclic, and tree-child, proving weak tree-childness. Rerooting on $B \to C$ makes A the parent of both reticulations C, F, so that admissible rooting is not tree-child. The blob contains exactly the two reticulations. In K_4, leaf 1 is adjacent to the triangle vertex B; in K_4', it is adjacent to the nontriangle vertex E. This labelled graph property is unchanged by isomorphism or triangle redirection. ☐

9.2 Exact Fourier overlap

Use the fourteen nonconstant four-leaf JC orbit coordinates A through O in the order fixed in appendix A. Both displayed-tree parameterizations satisfy

$$\begin{aligned} J-K-M+N&=0,\\ J-AH-BF+CE&=0,\\ GL-EN&=0,\\ L^2-BEH&=0,\\ BM-DL-B^2F+BCE&=0,\\ BEO-BGH-CEL+DEH&=0. \end{aligned} \tag{7}$$

On $BE\neq 0$, five equations reconstruct J,K,M,N,O from $A,\dots,H,L$, and the remaining equation is $L^2=BEH$. Hence the common localized ring is an irreducible eight-dimensional domain and is smooth on the physical locus because $\partial(L^2-BEH)/\partial H=-BE\neq 0$.

At one strict source point, the common first eight and final six coordinates are

$$(A,\dots,H)=\left(\frac{277}{8000},\frac{3}{40},\frac{67}{2400},\frac{127}{16000},\frac{27}{5000},\frac{81}{5000},\frac{153}{160000},\frac{9}{500}\right),$$
$$(J,\dots,O)=\left(\frac{27}{16000},\frac{261}{320000},\frac{27}{10000},\frac{27}{20000},\frac{153}{320000},\frac{183}{80000}\right).$$

The target parameters lie in $\mathbb{Q}(\beta)$, where β is the smaller root of

$$43337075\beta^2-36083110\beta+7336259=0,\qquad \frac{441}{1250}<\beta<\frac{3529}{10000}. \tag{8}$$

An exact isolating-interval calculation puts every edge multiplier and both inheritance probabilities strictly in $(0,1)$. The frozen verifier lists the parameters and checks all 256 Fourier coordinates in $\mathbb{Q}(\beta)$.

For K_4, an eight-coordinate gauge Jacobian determinant factors as

$$-\frac{P^3s^3Q^4tR^4u^3vS^3(s-1)^2(v-1)(v+1)^2}{16384},$$

and for K_4' the corresponding determinant is

$$-\frac{(P')^3x^2y^2z(R')^4w^4(S')^3(Q')^4(x-1)^2(z-1)(z+1)^3}{32768}.$$

Both are nonzero at the common point. Thus both images are eight-dimensional local submanifolds of the same smooth irreducible locus. The positive branch $L=\sqrt{BEH}$ and the reconstruction equations force the remaining coordinates from the first eight, so the images share a relative eight-dimensional neighborhood.

9.3 Leaf substitution

Replace a leaf a in both networks by the same cherry with new pendant multipliers $u,v\in(0,1)$. If P is the old tensor, then

$$\widetilde{P}(g_X,g_1,g_2)=P(g_X,g_1\oplus g_2)u^{\mathbf{1}[g_1\neq 0]}v^{\mathbf{1}[g_2\neq 0]}. \tag{9}$$

For nonzero h,

$$uv = \widetilde{P}(0,h,h), \qquad \frac{u}{v} = \frac{\widetilde{P}(g_X,h,0)}{\widetilde{P}(g_X,0,h)},$$

where g_X has total h. Strict positivity gives a unique positive analytic inverse for u, v and then for every coordinate of P. Cherry substitution therefore adds exactly two dimensions and preserves a common regular open region.

Proof of theorem 1.3. The four-leaf case is proposition 9.1 and the exact overlap above. Repeatedly substitute a cherry in the same corresponding leaf other than leaf 1 in both networks. The model dimension becomes $8+2(n-4) = 2n$. The displayed tree-child rootings extend, the original omnian witness remains, and leaf 1 stays adjacent to a triangle vertex on one side and a nontriangle vertex on the other. Thus the pair remains in $W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$, nonisomorphic, and non-T-equivalent for every $n \geq 4$. $\square$

10 Biological consequence and scope

Within the standard strongly tree-child level-2 class, exact infinite-data JC observations recover the bridge tree, every labelled descendant attachment, every cycle/theta core, and every port order. The only generic topological ambiguity is which vertex of an ordinary triangle carries the reticulation arrowheads. Physical bridge multipliers are not recovered by the topology theorem, and compatible triangle orientations need not realize the same distribution outside their common germ.

Strong tree-childness is essential. Immediately in the larger weakly tree-child class, the pair in section 9 moves a labelled component between a triangle vertex and a nontriangle vertex while preserving a full-dimensional regular JC region. Thus this is a genuine boundary in topological observability, not merely a change in root-presentation multiplicity.

11 Reproducibility and proof status

The release contains three deterministic entry points: a quick theorem crosswalk, a full replay of all committed exact certificates, and a bounded regeneration from primitive graph encodings. The primary and clean-room implementations share no graph canonicalizer, displayed-switching engine, descendant-mask code, separator selector, or rank routine at the load-bearing gates. Mutation suites reject relation deletion, changed port matching, source–target reversal, polynomial reassignment, broken restoration/probe parentage, reciprocal-only bridge gauges, and false physical-parameter recovery.

Historical failed certificates are preserved but excluded from every active manifest. These include the reciprocal-only bridge chart, schema keys that merged distinct rooted relations, ineffective mutations, and candidate Theta/Omega moves outside the locked class. The exact theorem-to-certificate crosswalk and all file hashes accompany the release.

A Four-leaf JC orbit coordinates

Represent the nonzero elements of G by $1, 2, 3$. The fourteen nonconstant JC orbit representatives are

symbol	tuple	name	symbol	tuple	name
A	0011	r_{34}	B	0101	r_{24}
C	0110	r_{23}	D	0123	t_{234}
E	1001	r_{14}	F	1010	r_{13}
G	1023	t_{134}	H	1100	r_{12}
J	1111	u_{1111}	K	1122	u_{1122}
L	1203	t_{124}	M	1212	u_{1212}
N	1221	u_{1221}	O	1230	t_{123}

The seven bounded-atlas invariant templates are stored as sparse lists

$$\sum_m c_m \prod_{j \in m} q_j,$$

with coordinate indices in this orbit convention. Their complete coefficient lists are released in `jc_root_spanning_atlas_data.py` and `seventh_invariant.json`. The theorem verifier first transports these inert coefficient lists through every ordered leaf map and then regenerates each pullback from the bound graph tensor. No result is accepted by looking up a topology identifier.

B Certificate crosswalk

Claim	Independent exact evidence
Standard convention, rooting, S_{TC}, primitive fixtures	`reviews/final_standard_convention/` and `reviews/root_probe/`.
Primitive cycle/theta theorem and repairs	`docs/GENERATOR_AND_SUPPORT_THEOREM.md`, regenerated by the primitive compiler and convention review.
Bridge fiber, slices, both cut inclusions, two-active crossing	`reviews/global_bridge/`; the clean-room verifier reconstructs all endpoint tensors, minors, kernels, and mutations.
Three-outgoing directed local universe	`reviews/n3_universe_generator/` independently generates the exact 10,826 raw and 10,466 merged relation multisets; `reviews/bounded_directed_relation_cleanroom/` independently checks their graphs, signs, transports, and 5,344 root bindings.
Four-outgoing θ_2 filter and root binding	`reviews/theta2_signature_gate/`: exact invariant deck, $18 + 42 + 132$ presentation theorem, transports, and mutations.
Fixed-full restoration forests	`reviews/final_hard_cover_cleanroom/` and `reviews/final_hard_cover_adversary/`.
Arbitrary subdivisions and ten-port bound	restoration terminals: compact-only clean-clone replay in `reviews/compact_probe_clean_clone_gate/`; all 62 direct residual anchors: `reviews/direct_anchor_probe_closure/`.
Ordinary triangle germ	`reviews/triangle_redirection_cleanroom/`; all 64 coordinates, physical ranks, graph quotient, and mutations.

Claim	Independent exact evidence
Weak-class sharpness	frozen package `../s_tc_jc_sharp_boundary/`; all 256 coordinates, two rank-eight minors, rooting census, and all-n inverse.
Whole-proof adversarial review	`reviews/final_outcome_p_referee_v2/`.

Data and code availability

All graph encodings, sparse invariants, exact certificates, independent verifiers, mutation transcripts, source, and deterministic reproduction commands accompany this manuscript in the project repository. No specialized phylogenetic-network classification package is required.

Use of generative AI

Generative AI systems assisted with exploratory algebra, implementation, drafting, and adversarial review. Several earlier automated claims failed adversarial audit, including a reciprocal-only bridge chart and candidate ambiguity moves outside the locked standard strong class; those claims are withdrawn and preserved only in quarantined history. The active theorem is supported by exact graph-derived certificates and independently written replays. The author is responsible for the mathematical statement, code, and final submission.

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